Questions and MCQ

Step 4B: Understand role of capital structure on capital budgeting

"Debt and Taxes"

Use the information for the question(s) below.

Rosewood Industries has EBIT of $450 million, interest expense of $175 million, and a corporate tax rate of 21%.

1) Rosewood's net income is closest to:
A) $450 million.
B) $217 million.
C) $290 million.
D) $95 million.
Answer: B
Explanation: Net income = (*EBIT* - Interest expense)(1 - τC)
= ($450 million - $175 million)(1 - .21) = $217.25 million

2) The total of Rosewood's net income and interest payments is closest to:
A) $270 million.
B) $392 million.
C) $290 million.
D) $450 million.
Answer: B
Explanation: Net income + Interest expense = (*EBIT* - Interest expense)(1 - τC)
= ($450 million - $175 million)(1 - .21) = $217.25 million + $175 million = $392.25 million

3) If Rosewood had no interest expense, its net income would be closest to:
A) $405 million.
B) $160 million.
C) $450 million.
D) $355 million.
Answer: D
Explanation: Net income = (*EBIT* - Interest expense)(1 - τC)
= ($450 Million - $0)(1 - .21) = $355.50 million

4) Assume that investors hold Google stock in retirement accounts that are free from personal taxes. Also assume that Google's current pre-tax WACC is 14% and the corporate tax rate is 21%. If Google were to issue sufficient debt at a pre-tax cost of 7% to give them a debt-to-value ratio of 0.5, then the Google's after-tax WACC would be closest to:
A) 10.4%.
B) 12.8%.
C) 13.0%.
D) 15.0%.
E) 16.0%.
Answer: B
Explanation: $rWACC = \frac{E}{E+D}re + \frac{D}{E+D}rd - \frac{D}{E+D}rdTc = 14\% - \frac{1}{2}(7\%)(21\%) = 13.265\%$

Note that $= \frac{E}{E+D}re + \frac{D}{E+D}rd$ is the pre-tax WACC.

5) Nielson Motors has no debt, and maintains a policy of holding $80 million in excess cash reserves, invested in risk-free treasury securities currently yielding 3%. If Nielson has a 21% corporate tax rate, the cost of permanently maintaining this $80 million reserve is closest to:
A) $0.85 million.
B) $16.8 million.
C) $24.0 million.
D) $28.0 million.
Answer: B
Explanation: Annual interest tax shield = Amount × Interest × Tax rate = $80 million × 3% × 21% = $0.504 million
PV Tax shield $= \frac{\text{Amount} \times \text{interest} \times \text{tax rate}}{\text{discount rate}} = \frac{\$80 \times 3\% \times 35\%}{3\%} = \$16.8$ million

6) Taggart Transcontinental currently has no debt and an equity cost of capital of 16%. Suppose that Taggart decides to increase its leverage and maintain a market debt-to-value ratio of 1/3. Suppose Taggart's debt cost of capital is 9% and its corporate tax rate is 21%. Assuming that Taggart's pre-tax WACC remains constant, then with the addition of leverage its effective after-tax WACC will be closest to:
A) 12.9%.
B) 13.0%.
C) 15.4%.
D) 16.0%.
Answer: C
Explanation: $rWACC = \frac{E}{E+D}re + \frac{D}{E+D}rd - \frac{D}{E+D}rdTe = 16\% - \frac{1}{3}(9\%)(21\%) = 15.37\%$

Note that $= \frac{E}{E+D}re + \frac{D}{E+D}rd$ is the pre-tax WACC.

7) Which of the following statements is FALSE?
A) Given a 21% corporate tax rate, for every $1 in new permanent debt that the firm issues, the value of the firm increases by $0.79.
B) The firm's marginal tax rate may fluctuate due to changes in the tax code and changes in the firm's income bracket.
C) Many large firms have a policy of maintaining a certain amount of debt on their balance sheets.
D) Typically, the level of future interest payments varies due to changes the firm makes in the amount of debt outstanding, changes in the interest rate on that debt, and the risk that the firm may default and fail to make an interest payment.
Answer: A
Explanation: Given a 21% corporate tax rate, for every $1 in new permanent debt that the firm issues, the value of the firm increases by $0.21.

8) Which of the following equations is INCORRECT?
A) $VL = VU + \frac{\tau_c D r_D}{r_D}$
B) $VL = VU + \tau c D$
C) $rwacc = \frac{E}{E + D} rE + \frac{D}{E + D} rD - \frac{D}{E + D} rD\tau c$
D) $rwacc = \frac{E}{E + D} rE + \frac{D}{E + D} rD(1 + \tau c)$
Answer: D
Explanation: $rwacc = \frac{E}{E + D} rE + \frac{D}{E + D} rD(1 - \tau c)$

Use the information for the question(s) below.

Flagstaff Enterprises expected to have free cash flow in the coming year of $8 million, and this free cash flow is expected to grow at a rate of 3% per year thereafter. Flagstaff has an equity cost of capital of 13%, a debt cost of capital of 7%, and it has a 35% corporate tax rate.

9) If Flagstaff maintains a .5 debt to equity ratio, then Flagstaff's pre-tax WACC is closest to:
A) 10.5%.
B) 11.0%.
C) 9.0%.
D) 10.0%.
Answer: B
Explanation: $rwacc = \frac{E}{E + D} rE + \frac{D}{E + D} rD$
$rwacc = \frac{1}{1 + .5}.13 + \frac{.5}{1 + .5}.07 = .11$

10) If Flagstaff currently maintains a .5 debt to equity ratio, then the value of Flagstaff as an all-equity firm would be closest to:
A) $80 million.
B) $100 million.
C) $73 million.
D) $115 million.
Answer: B
Explanation: $rwacc = \frac{E}{E+D} rE + \frac{D}{E+D} rD$ (Pre tax)

$rwacc = \frac{1}{1+.5}.13 + \frac{.5}{1+.5}.07 = .11$

$VU = \frac{FCF}{r_E - g} = \frac{\$8}{.11 - .03}$ million = $100 million

Use the information for the question(s) below.

LCMS Industries has $70 million in debt outstanding. The firm will pay only interest on this debt (the debt is perpetual). LCMS' corporate tax rate is 21% and the firm pays a rate of 8% interest on its debt.

11) Assuming that the risk of the tax shield is only 6% even though the debt pays 8%, then the present value of LCMS' interest tax shield is closest to:
A) $24.5 million.
B) $18 million.
C) $19.6 million.
D) $20.0 million.
Answer: C
Explanation: *PV* of Tax shield = debt × τC × *rD*/*RD*2 = $70M × .21 × .08/.06 = $19.60 million

12) Which of the following statements regarding recapitalizations is FALSE?
A) With a recapitalization, even though leverage reduces the total value of equity, shareholders capture the benefits of the interest tax shield up front.
B) The share price always rises after the completion of the recapitalization.
C) Leveraged recaps were especially popular in the mid- to late-1980s, when many firms found that these transactions could reduce their tax payments.
D) When a firm makes a significant change to its capital structure, the transaction is called a recapitalization.
Answer: B
Explanation: This is only always true if we ignore the other potential side effects of leverage, such as the costs of financial distress.

Use the information for the question(s) below.

Shepard Industries expects free cash flow of $10 million each year. Shepard's corporate tax rate is 21%, and its unlevered cost of equity is 10%. The firm also has outstanding debt of $40 million and it expects to maintain amount of debt permanently.

13) The value of Shepard Industries without leverage is closest to:
A) $114 million.
B) $50 million.
C) $100 million.
D) $64 million.
Answer: C
Explanation: $VU = FCF/rE$ = $10M/.10 = $100M

14) The value of Shepard Industries with leverage is closest to:
A) $64 million.
B) $100 million.
C) $135 million.
D) $108.4 million.
Answer: D
Explanation: $VU = FCF/rE$ = $10M/.10 = $100M
PV of Tax shield = $40 × .21 = $8.4M
$VL = VU + PV$ of tax shield = $100M + $8.4 = $108.4M

Use the table for the question(s) below.

Consider the following historical top federal tax rates in the United States:

Personal Tax Rates

Year	Corporate Tax Rate	Interest Income	Dividends	Capital Gains
2000	35%	40%	40%	20%
2005	35%	35%	15%	15%

15) In 2000, the effective tax rate for debt holders was closest to:
A) 61%.
B) 52%.
C) 64%.
D) 40%.
Answer: D
Explanation: Since there is no corporate tax on interest payments, the interest rate is simply the personal rate on interest income which is 40%.

16) In 2000, assuming an average dividend payout ratio of 50%, the effective tax rate for equity holders was closest to:
A) 69%.
B) 65%.
C) 55%.
D) 30%.
Answer: C
Explanation: The average personal tax rate on equity is (40% + 20%)/2 = 30%
So, the effective tax rate = 1 - (1 - τc)(1 - τe) = 1 - (1 - .35)(1 - .30) = .545

17) In 2000, assuming an average dividend payout ratio of 50%, the effective tax advantage for debt (t*) was closest to:
A) 40%.
B) 24%.
C) 30%.
D) 18%.
Answer: B
Explanation: The average personal tax rate on equity is (40% + 20%)/2 = 30%

$$\tau^* = \frac{\left(1 - \tau_i\right) - \left(1 - \tau_c\right)\left(1 - \tau_e\right)}{\left(1 - \tau_i\right)} = \frac{(1 - .40) - (1 - .35)(1 - .30)}{(1 - .40)} = 0.241667$$

18) Which of the following statements is FALSE?
A) Even though firms have not issued new equity, the market value of equity has risen over time as firms have grown.
B) While firms seem to prefer debt when raising external funds, not all investment is externally funded.
C) To receive the full tax benefits of leverage a firm needs to use 100% debt financing.
D) If bankruptcy is costly, these costs might offset the tax advantages of debt financing.
Answer: C
Explanation: To receive the full tax benefits of leverage a firm needs to be paying taxes.